A-A-51135A
April 21, 1986
SUPERSEDING
A-A-51135
October 23, 1984

COMMERCIAL ITEM DESCRIPTION

HOLE SAWS, HOLE SAW ARBORS, AND HOLE SAW DRIVE PLATES

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification GGG-S-66C.

1. SCOPE

1.1 Scope. This description covers hole saws, hole saw arbors, and hole saw drive plates of the standard, heavy duty, commercial type, which are used in a production environment.

1.2 Classification. This description covers hole saws of the following types, styles, sizes, and classes. The size is determined by the outside diameter (OD) in inches of the hole saw. This description also covers hole saw arbors, type III, styles V, W, and X; type IV, styles X and Y; type V, styles Y and Z; and hole saw drive plates, type VI, styles A and B. The type, style, size, and class hole saw, the type and style hole saw arbor, and the type and style hole saw drive plate shall be as specified (see 6.2).

Type I - Hole saws for use with types III, IV, and V arbors.

Type II - Hole saws with an integral arbor.

Style A - Hole saws with high speed steel teeth.
Style B - Hole saws with carbide tipped teeth.
Style C - Hole saws with an abrasive grit cutting edge.

AMSC N/A FSC 3455

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Size 9 - 9/16" OD	Size 26 - 1 5/8" OD	Size 50 - 3 1/8" OD
Size 10 - 5/8" OD	Size 27 - 1 11/16" OD	Size 52 - 3 1/4" OD
Size 11 - 11/16" OD	Size 28 - 1 3/4" OD	SIze 54 - 3 3/8" OD
Size 12 - 3/4" OD	Size 29 - 1 13/16" OD	Size 56 - 3 1/2" OD
Size 13 - 13/16" OD	Size 30 - 1 7/8" OD	Size 58 - 3 5/8" OD
Size 14 - 7/8" OD	Size 32 - 2" OD	Size 60 - 3 3/4" OD
Size 15 - 15/16" OD	Size 33 - 2 1/16" OD	Size 62 - 3 7/8" OD
Size 16 - 1" OD	Size 34 - 2 1/8" OD	Size 64 - 4" OD
Size 17 - 1 1/16" OD	Size 36 - 2 1/4" OD	Size 66 - 4 1/8" OD
Size 18 - 1 1/8" OD	Size 37 - 2 5/16" OD	Size 68 - 4 1/4" OD
Size 19 - 1 3/16" OD	Size 38 - 2 3/8" OD	Size 70 - 4 3/8" OD
Size 20 - 1 1/4" OD	Size 40 - 2 1/2" OD	Size 72 - 4 1/2" OD
Size 21 - 1 5/16" OD	Size 41 - 2 9/16" OD	Size 76 - 4 3/4" OD
Size 22 - 1 3/8" OD	Size 42 - 2 5/8" OD	Size 80 - 5" OD
Size 23 - 1 7/16" OD	Size 44 - 2 3/4" OD	Size 88 - 5 1/2" OD
Size 24 - 1 1/2" OD	Size 46 - 2 7/8" OD	Size 92 - 5 3/4" OD
Size 25 - 1 9/16" OD	Size 48 - 3" OD	Size 96 - 6" OD

Class a - Hole saws with a cutting depth of not less than 1 1/8 inches and not greater than 1 11/16 inches.

Class b - Hole saws with a cutting depth of not less than 1 3/4 inches and not greater than 2 1/2 inches.

Type III - Standard arbors for use with type I hole saws.

Style V - Arbors with a 1/4 inch shank used to drive sizes 9 through 19 hole saws.

Style W - Arbors with a 7/16 inch shank used to drive sizes 9 through 19 hole saws.

Style X - Arbors with a 7/16 inch shank used to drive sizes 20 through 96 hole saws.

Type IV - Quick disconnect arbors for use with type I hole saws.

Style X - Arbors with a 7/16 inch shank used to drive sizes 20 through 96 hole saws.

Style Y - Arbors with a 5/8 inch shank used to drive sizes 20 through 96 hole saws.

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Type V - Heavy duty arbors for use with type I hole saws.

Style Y - Arbors with a 5/8 inch shank used to drive sizes 20 through 96 hole saws.
Style Z - Arbors with a number 3 Morse taper shank and a permanently attached mounting plate used to drive sizes 48 through 96 hole saws.

Type VI - Drive plates for use with type I hole saws and style X and Y arbors.

Style A - For use with sizes 48 through 66 hole saws.
Style B - For use with sizes 68 through 96 hole saws.

2. APPLICABLE DOCUMENTS

2.1 Documents. The following documents of the issues in effect on the date of invitation for bids or request for proposals form a part of this description to the extent specified herein.

Federal Specification PPP-T-1150	- Tool and Tool Accessories for Power Driven, Metal and Woodworking Machinery, Packaging and Packing of.
Federal Standard FED-STD-H28	- Screw Thread Standards for Federal Services.
Federal Standard FED-STD-376	- Preferred Metric Units for General Use by the Federal Government.
American National Standards Institute, Standard ANSI/ASME B94.54	- Specifications for Hole Saws, Hole Saw Arbors, and Hole Saw Accessories.
American Society for Testing and Materials ASTM A 48	- Specification for Gray Iron Castings.
American Society for Testing and Materials ASTM D 3951	- Standard Practice for Commercial Packaging.
American Society for Testing and Materials ASTM F 711	- Specification for Fiberglass Reinforced Plastic (FRP) Rod and Tube Used in Line Tools.
American Society for Testing and Materials ASTM A 568	- Steel, Sheet, Carbon, and High-Strength, Low-Alloy, Hot-Rolled and Cold-Rolled, General Requirements for.

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2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. SALIENT CHARACTERISTICS

3.1 Design. The hole saws, hole saw arbors, and hole saw drive plates shall be new and of the manufacturer's commercial design in effect on the date of the solicitation. The hole saws, hole saw arbors, and hole saw drive plates shall meet the requirements specified herein and shall conform to the requirements of ANSI/ASME B94.54.

3.2 Materials. Materials used in the manufacture of hole saws, hole saw arbors, and hole saw drive plates shall be of a quality necessary to produce a product that conforms to the requirements specified herein.

3.3 Measurement system. All dimensions in this description are given in US units. These measurements may be converted to SI units through the use of conversion factors and methods specified in FED-STD-376.

3.4 Threads. All threaded parts shall conform to FED-STD-H28.

3.5 Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing employees to undue hazards from the recovered materials.

3.6 Hole saw design. Hole saws shall consist of a cutting edge, a body, and a cap. Hole saws shall rotate in a clockwise direction as viewed from the driving end.

3.6.1 Cutting edge. The cutting edge shall be permanently attached to the hole saw body and shall conform to style A, B, or C of this description.

3.6.1.1 Tooth set. Tooth set shall be alternate or raker.

3.6.1.1.1 Alternate set. The alternate tooth set shall have the cutting teeth set alternately one to the left, and one to the right throughout the circumference of the cutting edge.

3.6.1.1.2 Raker set. The raker tooth set shall have the cutting teeth set one to the left, one to the right, and one straight throughout the circumference of the cutting edge.

3.6.1.2 Pitch (number of teeth per inch). The pitch for hole saws shall be the manufacturer's standard pitch for hole saws furnished to the commercial market.

3.6.1.3 Tooth hardness. Hole saws shall have a tooth hardness of not less than 62 Rockwell C and not greater than 68 Rockwell C.

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3.6.2 Body. The hole saw body shall be manufactured of steel, shall be permanently attached to the hole saw cap, and shall contain core removal slots or holes.

3.6.2.1 Body thickness. The body thickness for hole saws shall be not less than 0.038 inch and not greater than 0.056 inch.

3.6.2.2 Body hardness. The body hardness of the hole saws shall be not greater than 52 Rockwell C.

3.6.3 Cap. The hole saw cap shall provide a means of driving the hole saw through the use of a separate arbor, an integral arbor, a mounting plate with an integral arbor, or a drive plate with a separate arbor.

3.6.3.1 Caps for type I, sizes 9 through 19 hole saws. Caps for type I, sizes 9 through 19 hole saws shall have a 1/2-20 internally threaded hole through the axis of the cap.

3.6.3.2 Caps for type I, sizes 20 through 46 hole saws. Caps for type I, sizes 20 through 46 hole saws shall have a 5/8-18 internally threaded hole through the axis of the cap, and not less than two 17/64 inch diameter drive holes located 180 degrees apart on a one inch diameter circle concentric with the center point of the cap.

3.6.3.3 Caps for type I, sizes 48 through 66 hole saws. Caps for type I, sizes 48 through 66 hole saws shall have a 5/8-18 internally threaded hole through the axis of the cap, not less than two 17/64 inch diameter drive holes located 180 degrees apart on a one inch diameter circle concentric with the center point of the cap, and not less than two 21/64 inch diameter drive holes located 180 degrees apart on a two inch diameter circle concentric with the center point of the cap.

3.6.3.4 Caps for type I, sizes 68 through 96 hole saws. Caps for type I, sizes 68 through 96 hole saws shall have a 5/8-18 internally threaded hole through the axis of the cap, not less than two 17/64 inch diameter drive holes located 180 degrees apart on a one inch diameter circle concentric with the center point of the cap, and not less than three 21/64 inch diameter drive holes located 120 degrees apart on a 3-3/8 inch diameter circle concentric with the center point of the cap.

3.6.3.5 Caps for type II hole saws. Caps for type II hole saws shall have a 7/16 inch hexagonal shank permanently attached to the center point of the cap. Shanks for type II hole saws shall have provisions to receive and secure a 1/4 inch pilot drill.

3.6.4 Performance. Each hole saw shall be capable of meeting the sawing requirements specified in table I. The test material shall be 1/4 inch thick 1018 cold finished steel conforming to ASTM A 568 for style A hole saws, 1/2 inch thick class 35 gray iron conforming to ASTM 48 for style B hole saws, and 1/2 inch thick fiberglass conforming to ASTM F 711 for style C hole saws. The diameter of each hole cut shall be not greater than 0.060 inch larger than the hole saw diameter for any diametrical measurement.

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TABLE I. Sawing test requirements.

Hole saw size	Number of holes to be sawed	Maximum time allowed for each hole cut
9 through 16	10	35 seconds
17 through 32	6	70 seconds
33 through 48	5	100 seconds
50 through 64	4	140 seconds
66 through 96	3	205 seconds

3.6.5 Hole saw concentricity and tooth crest runout. Hole saw concentricity and tooth crest runout shall not exceed the tolerances contained in table II.

TABLE II. Tolerances for hole saw concentricity and tooth crest runout.

Hole saw sizes	Concentricity of saw body (inches)	Radial tooth crest runout (inches)	
		Longitudinal seam	Total runout
9 through 30	0.030	0.010	0.030
32 through 62	0.040	0.010	0.040
64 through 96	0.060	0.010	0.060

3.7 Arbors.

3.7.1 Standard arbors. Standard arbors shall be manufactured of steel.

3.7.1.1 Type III, style V standard arbors. Type III, style V standard arbors shall have a 1/4 inch hexagonal shank attached to the hole saw cap by means of 1/2-20 external threads. The arbor body diameter shall be of sufficient section to resist breakage with provisions to receive and secure a 1/4 inch pilot drill. The pilot drill may extend through the arbor in lieu of the hexagonal shank.

3.7.1.2 Type III, style W standard arbors. Type III, style W standard arbors shall have a 7/16 inch hexagonal shank attached to the hole saw cap by means of 1/2-20 external threads. The arbor body diameter shall be of sufficient section to resist breakage with provisions to receive and secure a 1/4 inch pilot drill.

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3.7.1.3 Type III, style X standard arbors. Type III, style X standard arbors shall have a 7/16 inch hexagonal shank attached to the hole saw cap by means of 5/8-18 external threads. The arbor body diameter shall be of sufficient section to resist breakage with provisions to receive and secure a 1/4 inch pilot drill.

3.7.2 Quick disconnect arbors. Quick disconnect arbors shall be manufactured of steel and provide for interchangeability of hole saws without the use of mechanical tools. Quick disconnect arbors shall utilize retractable drive pins to provide positive drive to the hole saw and prevent jamming between the arbor base and hole saw cap. Quick disconnect arbors shall conform to the following.

3.7.2.1 Type IV, style X quick disconnect arbors. Type IV, style X quick disconnect arbors shall have a 7/16 inch hexagonal shank attached to the hole saw cap by means of 5/8-18 external threads. The arbor body diameter shall be of sufficient section to resist breakage with provisions to receive and secure a 1/4 inch pilot drill. The arbor shall have two 1/4 inch drive pins located 180 degrees apart on a one inch diameter circle concentric with the vertical axis of the arbor. Drive pins shall be of sufficient length to extend into the hole saw cap not less than 1/4 inch and not greater than 3/8 inch.

3.7.2.2 Type IV, style Y quick disconnect arbors. Type IV, style Y quick disconnect arbors shall have a 5/8 inch hexagonal shank attached to the hole saw cap by means of 5/8-18 external threads. The arbor body diameter shall be of sufficient section to resist breakage with provisions to receive and secure a 1/4 inch pilot drill. The arbor shall have two 1/4 inch drive pins located 180 degrees apart on a one inch diameter circle concentric with the vertical axis of the arbor. Drive pins shall be of sufficient length to extend into the hole saw cap not less than 1/4 inch and not greater than 3/8 inch.

3.7.3 Heavy duty arbors. Heavy duty arbors shall be manufactured of steel.

3.7.3.1 Type V, style Y heavy duty arbors. Type V, style Y heavy duty arbors shall have a 5/8 inch hexagonal shank attached to the hole saw cap by means of 5/8-18 external threads. The arbor body diameter shall be of sufficient section to resist breakage with provisions to receive and secure a 1/4 inch pilot drill. The arbor shall have two removable drive pins located 180 degrees apart on a one inch diameter circle concentric with the vertical axis of the arbor. Drive pins shall provide a positive drive to the hole saw and prevent jamming between the arbor base and hole saw cap. Drive pins shall be of sufficient length to extend into the hole saw cap not less than 1/4 inch and not greater than 3/8 inch.

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3.7.3.2 Type V, style Z heavy duty arbors. Type V, style Z heavy duty arbors shall have a number 3 Morse taper shank integrally attached to a mounting plate. The mounting plate shall be not less than 3-15/16 inches and not greater than 4-1/8 inches in diameter, and not less than 3/8 inch and not greater than 1/2 inch thick. The mounting plate shall have not less than three 5/16-18 internally threaded holes located 120 degrees apart on a 3-3/8 inch diameter circle concentric with the center point of the plate, and not less than two 5/16-18 internally threaded holes located 180 degrees apart on a two inch diameter circle concentric with the center point of the plate. A 3/8 inch pilot pin shall be located at the vertical axis of the mounting plate.

3.8 Drive plates. Drive plates shall be manufactured of steel and conform to the following requirements for the style to be furnished.

3.8.1 Type VI, style A drive plates. Type VI, style A drive plates shall be not less than 2-1/2 inches and not greater than 3 inches in diameter, and not less than 3/8 inch and not greater than 1/2 inch thick. The drive plate shall have a 5/8-18 internally threaded hole through the center point of the plate, not less than two 17/64 inch diameter drive holes located 180 degrees apart on a one inch diameter circle concentric with the center point of the plate, and not less than two 5/16-18 internally threaded holes located 180 degrees apart on a two inch diameter circle concentric with the center point of the plate. Two 5/16-18 set screws 1/2 inch in length shall be supplied with each drive plate ordered. These drive plates are used with styles X and Y arbors to drive sizes 48 through 66 hole saws.

3.8.2 Type VI, style B drive plates. Type VI, style B drive plates shall be not less than 4 inches and not greater than 4-1/4 inches in diameter, and not less than 3/8 inch and not greater than 1/2 inch thick. The drive plates shall have a 5/8-18 internally threaded hole through the center point of the plate, not less than two 17/64 inch diameter drive holes located 180 degrees apart on a one inch diameter circle concentric with the center point of the plate, and not less than three 5/16-18 internally threaded holes located 120 degrees apart on a 3-3/8 inch diameter circle concentric with the center point of the plate. Three 5/16-18 set screws 1/2 inch in length shall be supplied with each drive plate ordered. These drive plates are used with styles X and Y arbors to drive sizes 68 through 96 hole saws.

3.9 Pilot drills. When specified (see 6.2), 1/4 inch pilot drills shall be furnished. Pilot drills shall be manufactured of high speed steel with a hardness of not less than 62 Rockwell C and are for use with all arbors except style Z. The shank of the pilot drill shall be flat or contain not less than one notch which will secure the pilot drill to the arbor through the use of not less than one set screw. After the pilot drill is secured to the arbor the drill point shall extend at least 1/8 inch beyond the cutting edge of the hole saw.

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3.10 Workmanship. The workmanship of the hole saws, hole saw arbors, and hole saw drive plates shall be of a quality equal to that of comparable products furnished to the commercial market.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection, examination, and test requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this description where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sample inspection. The supplier shall randomly select the number of samples in table III that is applicable to the quantity of items being purchased. Each sample item shall be inspected to determine conformance with the examination in 4.3 and the dimensional inspection in 4.4. Any sample failing to meet all requirements specified herein shall be considered defective. If the number of defective samples is equal to or less than the quantity specified under the "Acceptance number defective" column of table III, the supplier shall replace the defective samples with samples that meet all the requirements specified herein and the entire lot shall be considered acceptable. If the number of defective samples is equal to or greater than the quantity specified in the "Rejection number defective" column of table III, the entire lot shall be considered unacceptable. A lot is defined as all items of the same type, style, size, and class presented at one time for acceptance.

TABLE III. Sampling plan.

Quantity of items being purchased	Sample size	Acceptance number defective	Rejection number defective
2 to 8	2	0	1
9 to 15	3	0	1
16 to 25	5	0	1
26 to 50	8	1	2
51 to 90	13	1	2
91 to 150	20	2	3
151 to 280	32	3	4
281 to 500	50	5	6

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4.3 Examination. Each hole saw, hole saw arbor, and hole saw drive plate submitted for sample inspection shall be visually examined to determine conformance with all requirements in this description, except for the performance requirements in 3.6.4 and hardness requirements in 3.6.1.3 and 3.6.2.2.

4.4 Dimensional inspection. Each hole saw, hole saw arbor, or hole saw drive plate submitted for sample inspection shall be inspected to assure conformance with all dimensional and tolerance requirements specified herein.

4.5 Performance test. After the sample inspection has been completed, a performance test shall be performed on the nearest whole number to 10 percent of the hole saw samples. The performance test shall conform to the requirements contained in 3.6.4. The rotational speed, cutting force, type of machine and machining techniques shall be in accordance with standard industrial procedures. After the test is performed, the hole saw shall show no evidence of overheating or physical damage. If any of the hole saw samples fail to pass the performance test specified herein, the entire lot from which the samples were selected shall be rejected.

4.6 Hardness test. After the performance test has been completed, a hardness test shall be performed on the same hole saws which have been subjected to the performance test. If any of the hole saw samples fail to meet the hardness requirements in 3.6.1.3 and 3.6.2.2, the total lot from which the samples were selected shall be rejected. The hole saws subjected to the performance test and hardness test shall not be furnished as part of the contract order.

4.7 Packaging inspection. Packaging shall be inspected for conformance with section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified in the contract or purchase order, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2), preservation, packing, and marking shall be in accordance with Federal Specification PPP-T-1150, level as specified.

6. NOTES

6.1 Intended use. Hole saws covered by this description are intended to be used for sawing round holes in metal, plastic, wood, and other similar materials.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this description.
b. Type, style, size, class, and quantity required (see 1.2).
c. Pilot drills, if required, specify quantity required (see 3.9).
d. Packaging in accordance with PPP-T-1150, if required; specify level (see 5.1).

6.3 Cross reference. Items covered by Federal Specification GGG-S-66C and Commercial Item Description A-A-51135 are cross referenced to items covered by this commercial item description as follows:

GGG-S-66C	A-A-51135	A-A-51135A
Type I	Type I	Type I
Type II	Type II	Type II
		Type III (New type)
		Type IV (New type)
		Style A (New style)
		Style B (New style)
		Style C (New style)
	Style V (New style)	Style V
	Style W (New style)	Style W
	Style X (New style)	Style X
	Style Y (New style)	Style Y
	Style Z (New style)	Style Z
		Class a (New class)
		Class b (New class)
Sizes 9/16 through 6	Sizes 9/16 through 6	Sizes 9 through 96

MILITARY INTEREST:

Custodians

Army-AL
Air Force - 99

Review Activities

Air Force - 84
DLA-GS

User Activities

Navy-MC, SH

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

Preparing Activity:

DLA - IP

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